

This Program Prepares You for – High-demand Job Roles

Current role and upgraded role after IIT Roorkee program

DATA SCIENTIST

Skills

Python, R, SQL, Basic Statistics, Data Visualization

₹4 - 12 LPA



AI ENGINEER WITH GENAI SPECIALIZATION

Upgraded Skills from IIT Roorkee

LLM fine-tuning, Prompt Engineering, LangChain, AutoGen, CrewAI, Agentic Workflow Design, RAG, Vector Databases

₹18 - 35 LPA



MACHINE LEARNING ENGINEER

Current Skills

Python, Basic TensorFlow/PyTorch, Simple Model Training

₹5 - 10 LPA



GENERATIVE AI ENGINEER

Upgraded Skills from IIT Roorkee

Advanced GenAI, LLMs (GPT-4, Cohere), Hugging Face, Diffusion Models, GANs, VAEs, Cloud ML Platforms (AWS SageMaker, Azure ML)

₹20 - 40 LPA



DATA ENGINEER

Current Skills

SQL, Basic ETL, Python, Simple Data Pipelines

₹4 - 8 LPA



AI DATA ENGINEER

Upgraded Skills from IIT Roorkee

Unstructured Data Processing, AI-driven ETL, Real-time Data Streaming, Cloud-native Pipelines (AWS, Azure, GCP), Vector Databases

₹15 - 25 LPA



MLOPS ENGINEER

Current Skills

Basic Docker, Simple CI/CD, Model Deployment

₹6 - 10 LPA



AI SYSTEMS ENGINEER WITH AIOPS

Upgraded Skills from IIT Roorkee

AIops, Multi-agent Orchestration, Advanced MLOps, Kubernetes, Prometheus, Grafana, Seldon, Evidently AI, Cloud Security

₹22 - 45 LPA



DATA ANALYST**Skills**

SQL, Excel, Basic Python, Tableau/
PowerBI

₹4 - 8 LPA

**AI PRODUCT ANALYST****Upgraded Skills from IIT Roorkee**

Comprehensive AI/ML understanding, Business
Impact Analysis, Basic GenAI applications, Data-
driven insights with AI tools

₹18 - 30 LPA

**BUSINESS INTELLIGENCE DEVELOPER****Skills**

SQL, Basic ETL, Power BI/Tableau,
Data Visualization

₹4 - 7 LPA

**CLOUD AI SOLUTIONS DEVELOPER****Upgraded Skills from IIT Roorkee**

AI/ML integration, Cloud platforms, Basic system
design, Performance optimization, Serverless AI

₹20 - 35 LPA

**AI ENGINEER****Skills**

Python, Basic ML Algorithms, Simple
Neural Networks

₹5 - 12 LPA

**AGENTIC AI ENGINEER****Upgraded Skills from IIT Roorkee**

Advanced Agentic AI, Multi-agent Systems,
Semantic Kernel, Knowledge Base Integration,
Production AI Deployment

₹25 - 50 LPA

**CLOUD ENGINEER****Skills**

Basic AWS/Azure, Simple Deployments,
Basic Networking

₹4 - 8 LPA

**AGENTIC AI ENGINEER****Cloud AI Engineer**

Advanced Cloud AI Services, Serverless AI,
Auto-scaling ML Infrastructure, Cost Optimization,
AI-native cloud architecture

₹25 - 45 LPA





◆ Role Evolution for Freshers & 1+ Years of Experience

Make the shift from surface learning to engineering production-grade GenAI systems, hands-on, end-to-end, and built for the real world.

GENERATIVE AI ENGINEER

Skills Required

Advanced GenAI, LLMs (GPT-4, Cohere), Hugging Face, Diffusion Models, GANs, VAEs, Cloud ML Platforms (AWS SageMaker, Azure ML)

₹12 - 30 LPA

STARTING SALARY

AI ENGINEER

Skills Required

Python, Machine Learning basics, TensorFlow/PyTorch, LLM fine-tuning, Prompt Engineering, LangChain, AutoGen, CrewAI

₹12 - 25 LPA

STARTING SALARY

MACHINE LEARNING ENGINEER

Skills Required

Python, Basic TensorFlow/PyTorch, Model Training, Data Preprocessing, Feature Engineering, Model Evaluation

₹12 - 28 LPA

STARTING SALARY

DATA SCIENTIST

Skills Required

Statistical Analysis, Python, R, SQL, Data Visualization, Machine Learning fundamentals

₹12 - 22 LPA

STARTING SALARY

AIOPS ENGINEER

Skills Required

Basic Docker, Simple CI/CD, Model Deployment, Kubernetes basics, Prometheus, Grafana, Cloud platforms

₹15 - 35 LPA

STARTING SALARY

DATA ANALYST

Skills Required

SQL, Excel, Basic Python, Tableau/PowerBI, Statistical Analysis, Data Visualization

₹12 - 18 LPA

STARTING SALARY

AI PRODUCT ANALYST

Skills Required

Comprehensive AI/ML understanding, Business Impact Analysis, Data-driven insights, Product Management basics

₹15 - 30 LPA

STARTING SALARY

CLOUD AI ENGINEER

Skills Required

Basic AWS/Azure, Simple Deployments, Cloud Security, Serverless AI, Auto-scaling ML Infrastructure

₹18 - 35 LPA

STARTING SALARY

Curriculum

Module 1: Foundations of AI, ML, and Generative AI

Topics

- Understanding Supervised Machine Learning, Unsupervised Machine Learning, and Reinforcement Learning
- Data Manipulation for Machine Learning
- Exploratory Data Analysis for Machine Learning
- Essential Mathematics for Machine Learning and AI
- Definition of Generative AI
- Generative AI Applications
- Definition of LLMs

Module 2: Deep dive into Machine Learning Algorithms

Topics

- Regression Supervised Machine Learning
 - Linear Regression
 - Decision Tree Regressor
 - Random Forest Regressor
 - Overfitting, Underfitting, and Regularization
- Classification Supervised Machine Learning
 - Logistic Regression
 - Decision Tree Classifier
 - Random Forest Classifier
- Unsupervised Machine Learning
 - K-Means Clustering
 - Hierarchical Clustering
 - Dimensionality Reduction
 - Anomaly detection
- MLOps for Machine Learning model deployment

Module 3: Understanding Deep Learning

Topics

- Exploring Deep Learning
Understanding Neural Networks with TensorFlow
- Deep dive into Neural Networks with TensorFlow
- Artificial Neural Networks (ANN)
- Convolutional Neural Networks (CNN)
- Object detection using the YOLO framework
- Transfer Learning in CNN
- Recurrent Neural Networks (RNN) and LSTMs

Module 4: Exploring NLP

Topics

- Architectures of RNN - One to One, Many to Many, Many to One, etc
- Understanding Text for NLP
Tokenization, Stemming and Lemmatization, Stopwords, and Keywords in NLP
- Text Vectorization using TF-IDF, Bag of Words, and Word 2 Vec
- Use cases across domains in NLP

Module 5: Engaging with Generative AI

Topics

- How Generative AI works
- Text generation, Image generation
- Audio generation and video generation
- Using Hugging face to access models for text generation, image generation
- GANs
Types of GANs - ProGAN, SRGAN, CycleGAN
- Auto Encoders, Variational Auto Encoders (VAEs), Diffusion Models

Module 6: Understanding LLMs

Topics

- Definition of LLMs
LLM Use Cases
- Prompt Tuning
- Attention Mechanism
- Transformer Model and Architecture
- Encoder-Decoder arrangements
- Train and generate text using
- Encoder-decoder Architecture
- BERT for Transfer learning
- Leveraging multiple pretrained LLMs from Hugging Face
- Fine-tuning LLMs

Module 7: GenAI Application Development Prompting Techniques

Topics

- Introduction to Prompt Engineering
- Successful and Unsuccessful prompts
- Types of Prompting
- Introduction to Open AI, GPT , Open AI Playground
- Cost & latency considerations when calling APIs (OpenAI, Azure, AWS).
- Multimodal prompting for GPT 4
- Image generation using Open AI DALLE 3
- Prompt evaluation
- Implementing Agents and Chains
- Implementing zero-shot-react, conversational-react agents with LangChain
- Open AI Function (Tool Integration)
- Testing various LLMs with Prompt Engineering

Module 8: Synthetic Data and Datasets for LLMs

Topics

- Introduction to Synthetic Data
- Generating Synthetic Data
- Synthetic Data for LLMs
- Real-world Applications and Use Cases
- Hands-on generating and using Synthetic Data

Module 9: AI Embeddings & Retrieval

Topics

- Understanding AI Embeddings
- Advanced Retrieval Techniques
- Hugging Face Embeddings
- Vector Databases in AI
- CRUD operations with Vector Databases
- RAG - Retrieval Augmented Generation
- RAG solutions using Open AI models and Hugging face models
- Ethical Considerations in AI Embeddings
- Navigating AI Hallucinations, Drift, and Bias
- Embeddings in Real-world Applications
- Embeddings Optimization and Fine-tuning
- Embeddings Security and Privacy
- LLM Ops and model deployment best practices

Module 10: Understanding Agentic AI

Topics

- Agents, Agentic AI, and Multi-Agent Systems
- Agent Definition & Autonomy
Simple vs. Knowledge-Based Agents
- Reflex vs. Goal-Driven Agents
Microsoft AutoGen
- Agent Architecture (Perception, Decision, Action)
- Integrating Knowledge Bases (RAG, Domain Data)
- Measuring Performance (Success Rate, Resource Usage)
- Hierarchical Agent Planning
- Multi-Step Reasoning with LLM
- Memory & Long-Term Context
- Integrating Retrieval Augmentation in Agent Workflows
- Domain-Specific Knowledge & Dynamic Prompting

Module 11: Exploring Cloud Ecosystems

Topics

- Cloud Ecosystems
- Introduction to Cloud Ecosystem
Definitions
- Cloud characteristics
- Deployment models
- Leading Service providers (AWS, Google, Azure, etc.)
- Comparing AWS, Azure, and GCP core services for compute, storage, and AI/ML.
- Data Centres and their components
- Service (SaaS, IaaS, PaaS)
Issues & Challenges

Module 12: Advancement in Hypervisors

Topics

- Understanding hypervisors
Reference model
- Virtualisation characteristics
- Principles of hypervisor design Interfaces
- Types of hypervisors (type-1 and type-2)
- Differences between Type-1 and Type-2 hypervisors.
- Design methods of hypervisors (full virtualization, para virtualization, and hardware-assisted virtualization)
- Memory Virtualisation
- I/O virtualisation
- OS virtualization
- Comparative Analysis of hypervisors
- Understanding performance, requirements, and bottleneck

Module 13: Container Orchestration

Topics

- Understanding LLM Deployment Architectures
- Containerizing LLM Inference Services (e.g., using FastAPI + Docker)
- Managing GPU Workloads in Kubernetes
- Scaling LLM APIs with Kubernetes and Istio
- Optimizing Latency and Throughput for LLM Containers
- Secure Access and Rate Limiting for AI APIs
- CI/CD for LLM-Powered Microservices
- Monitoring and Logging for LLM Containers
- Model Versioning and Rollbacks
- Cost Optimization Strategies for LLM Inference in Production

Module 14: Cloud Security & Resilience

Topics

- Infrastructure security: Network-level security, Host-level security, Application-level security
- Data security and storage: Data privacy and security issues
Jurisdictional issues raised by data location
- Identity and access management
- Access control
- IAM, Key Management Services, and zero-trust architecture
- Trust, reputation, risk
Authentication in cloud computing
- Client access in the cloud
- Cloud contracting model
- Commercial and business considerations

Module 15: Understanding Cloud in context of Gen AI and LLMs

Topics

- EC2 Deep Dive and AMLs, EBS vs S3 vs EFS – Storage Solutions
- Load Balancing and Auto Scaling Basics
- Intro to Serverless: AWS Lambda
- Using AWS Bedrock for GenAI (including foundation models)
- Deploying Open-Source LLMs on EC2/EKS
- Fine-Tuning and Inference
- Pipelines on Cloud
- Cost & Performance
- Considerations for LLM Workloads
- SageMaker Pipelines for Model
- Training and Inference
- Model Versioning, A/B Testing, and Rollbacks
- Security and Compliance for GenAI in Production